

Nucleic acids

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First stage

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Nucleic acids

Nucleic acids are complex structures used to maintain genetic information.

Nucleotides are found primarily as the monomeric units comprising the major nucleic acids of the cell, DNA and RNA.

Deoxyribonucleic acid (DNA) serves as the master copy for most information in the cell.

Ribonucleic acid (RNA) are different types, they act to transform information from DNA to the most of the cell.

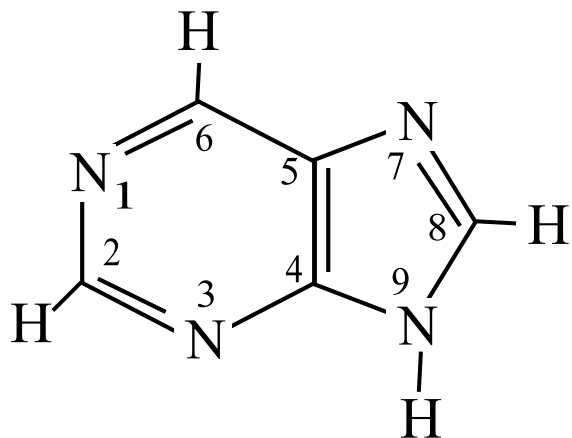
The primary structure of both DNA and RNA are similar, both are poly nucleotides, and each nucleotide composed of three units, nitrogenous base, a pentose sugar and phosphate group (at least one).

1-Nitrogenous bases (heterocyclic compounds)

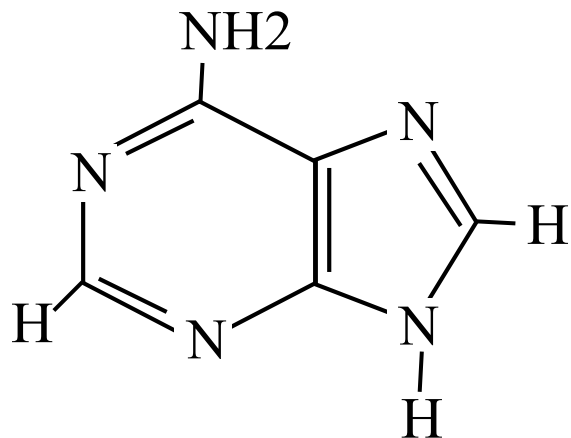
Heterocyclic compounds are ring compounds that contain carbon and non carbon (hetero) atoms.

There are five heterocyclic compounds that which are derivatives of either purine or pyrimidine constitute nucleic acids. Adenine and guanine are derivatives of purine while thymine, cytosine and uracil are derivatives of pyrimidine.

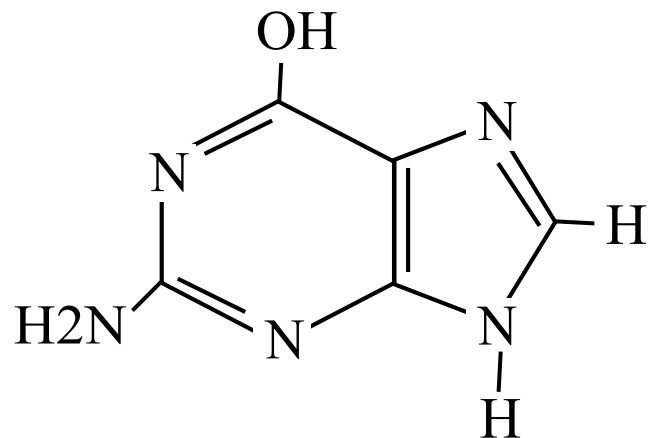
When these compounds are dissolved in water, they can form a basic solution, for this reason they are commonly called heterocyclic bases.



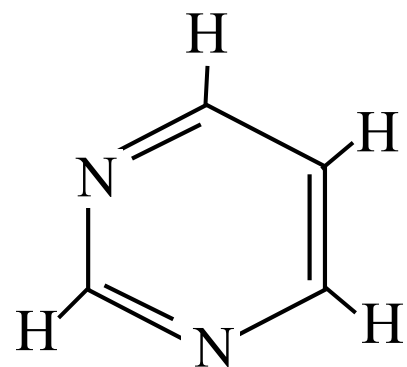
Purine (Parent compound)



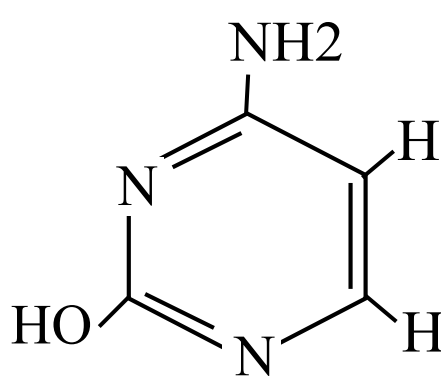
Adenine (A)



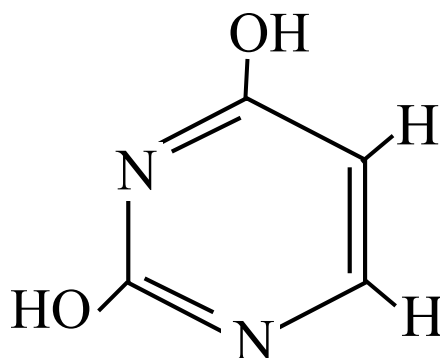
Guanine (G)



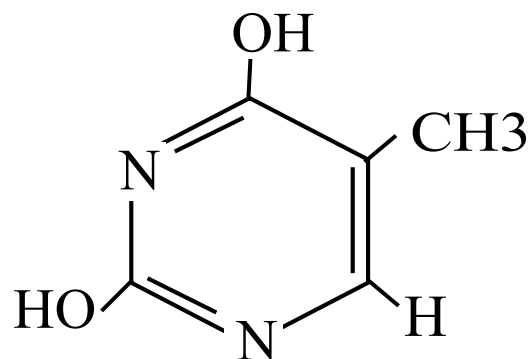
Pyrimidine
(Parent compound)



cytosine ©



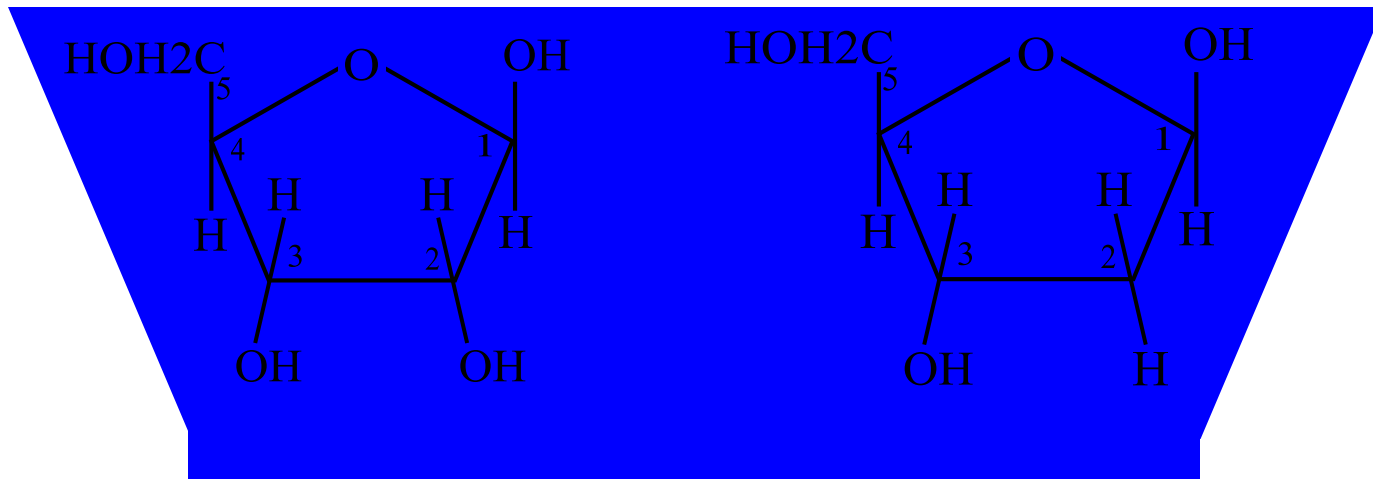
uracil (U)



thymine (T)

2- Sugars in nucleic acid (pentose sugar)

A nucleic acid will contain either ribose or deoxy ribose as its sugar. The sugar in RNA is ribose and that in DNA is deoxy ribose, lacking one oxygen atom (hence the name de-oxy).



Ribose

Deoxy ribose

3-Phosphate group (inorganic phosphate)

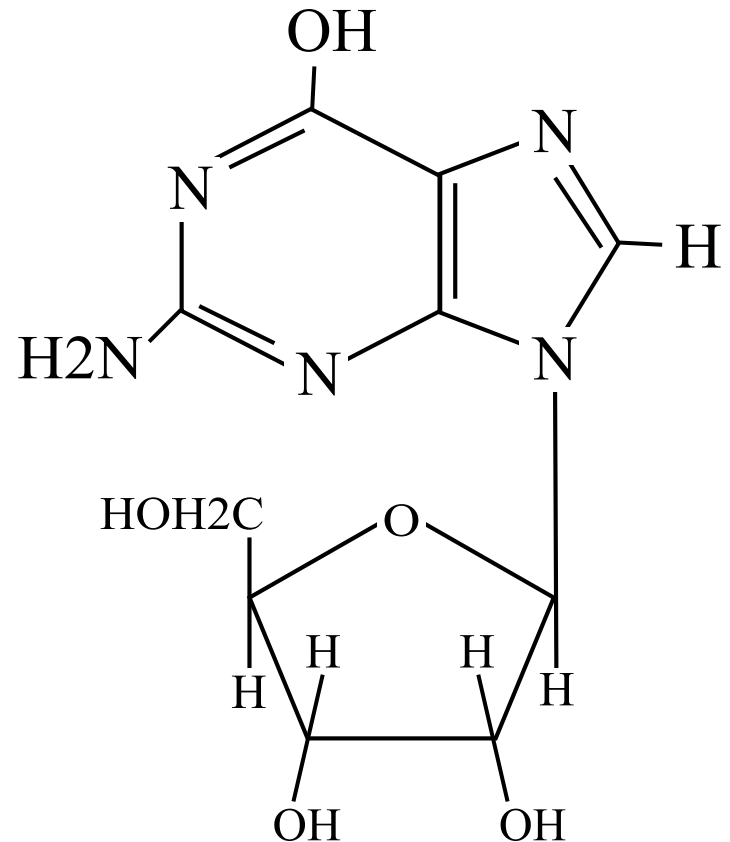
Each nucleotide contain one or two or three phosphate group.

Nucleosides

Each one of the heterocyclic bases can be combined with either ribose or deoxy ribose to form a nucleoside

Nucleosides:-

Structure of Guanosine



Names of nucleosides

Adenine + ribose = Adenosine

Guanine + ribose = Guanosine

Thymine + ribose = Thymidine

Cytosine + ribose = Cytidine

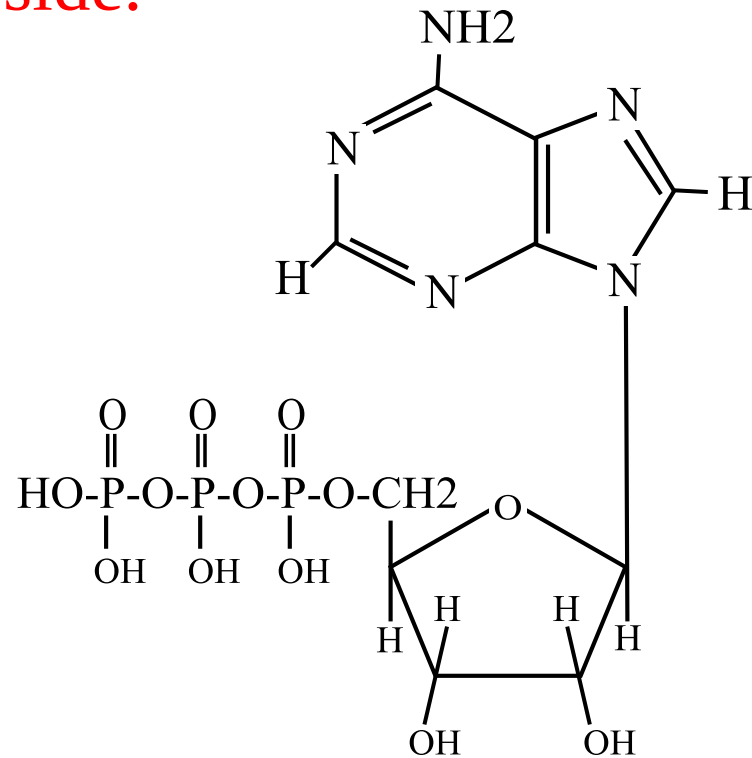
Uracil + ribose = Uracine

Nitrogen bases with deoxy ribose produce deoxy **adenosine**, deoxy **guanosine**, deoxy **thymidine**, deoxy **cytidine**, and deoxy **uracine**

Nucleotides

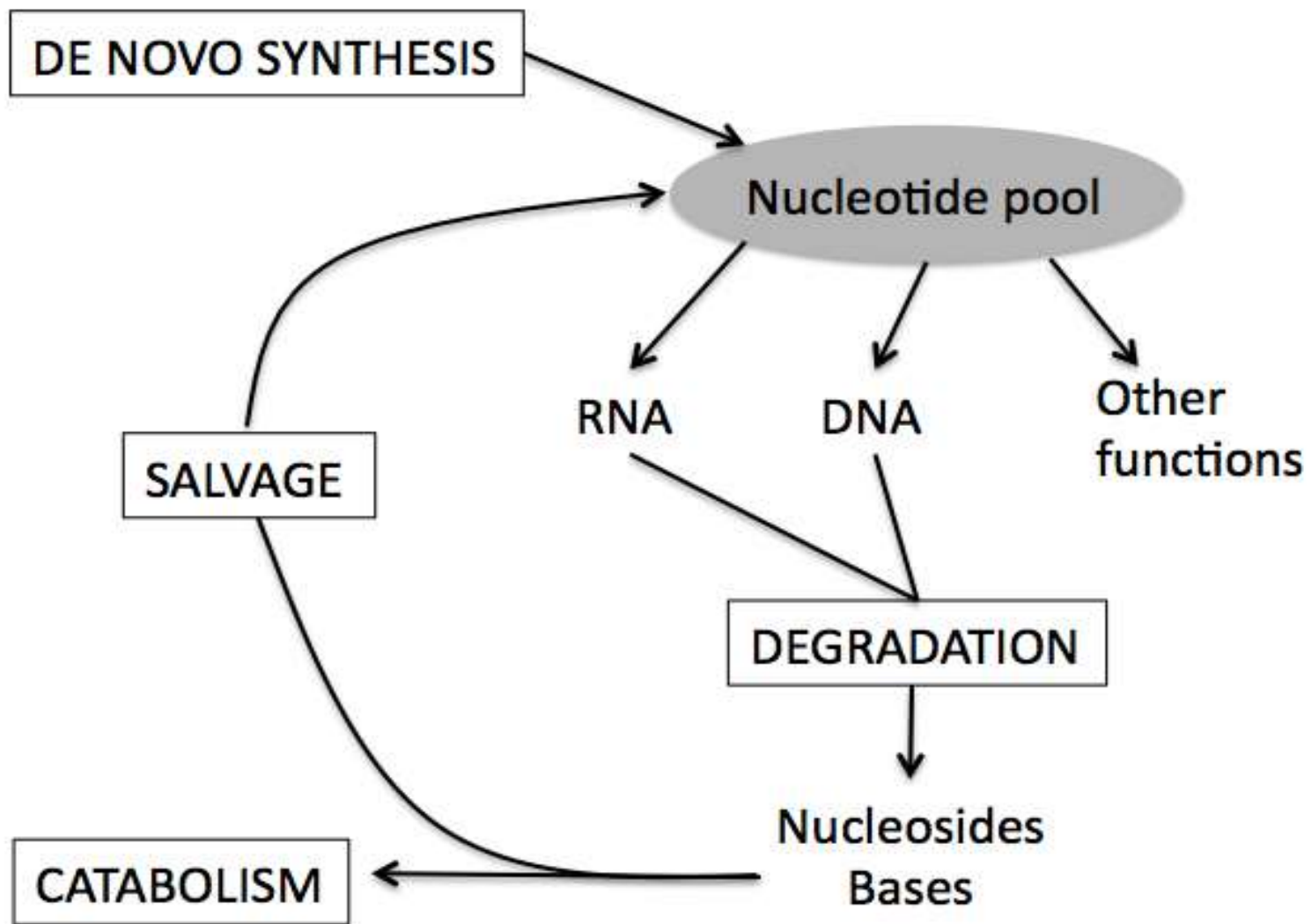
Nucleotides are esters of nucleosides. All naturally nucleotides contain the phosphate group attached to carbon No.5 of the pentose sugar of nucleoside.

Molecules with two or three phosphate groups are good energy donors, releasing energy with the transfer of phosphate groups. Nucleotides such as ATP (Adenosine tri-phosphate) and GTP (Guanosine tri-phosphate) are used not just for RNA or DNA synthesis, but also as energy donors for many cellular reactions.

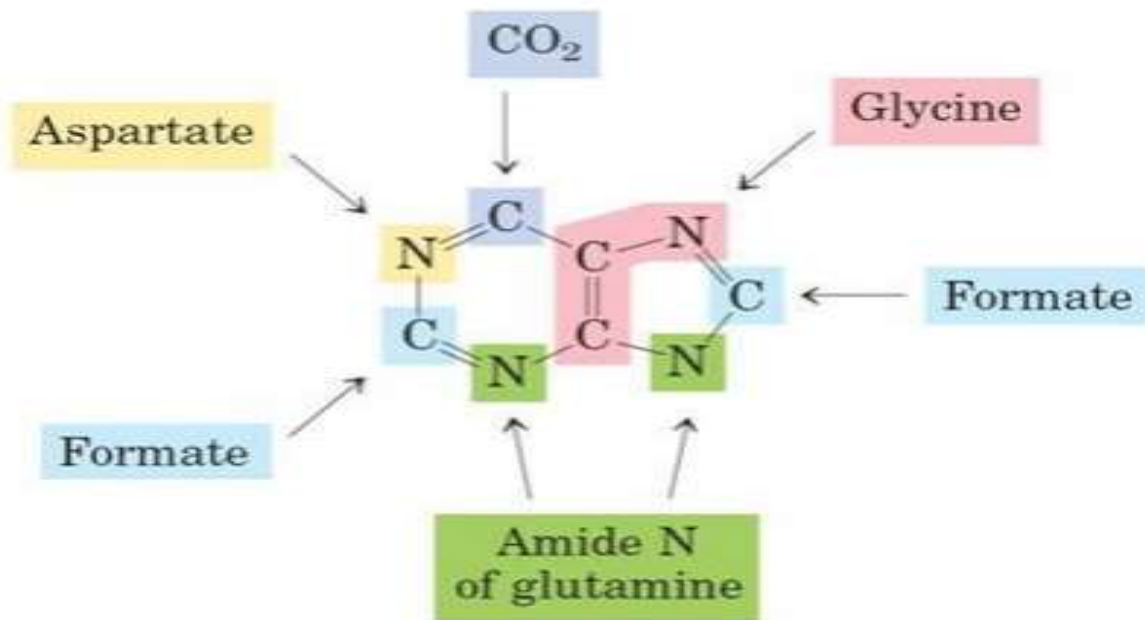


Biosynthesis of purine nucleotides

- The purine nucleotides are synthesized by most of the tissues.
- The major site is the liver. This pathway operates in the cytoplasm.
- The major pathway is denoted as *de novo synthesis*, because the purine ring is synthesized from different small components and from different sources.
- During *de novo* synthesis, purine ring is buildtup on a *ribose-5-phosphate* molecule.

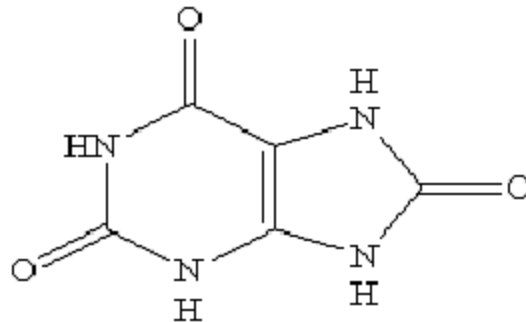


❑ The **purine** ring is constructed by a series of reactions that add the donated carbons and nitrogen to a preformed ribose 5-phosphate.



Degradation of purine nucleosides

- ❑ The end product of purine nucleotide catabolism is uric acid (Urate).
- ❑ This degradation is taking place mainly in the liver.
- ❑ Normal blood level of uric acid ranges from 3-7 mg/dl.
- ❑ Nucleic acid content is more in red meat.

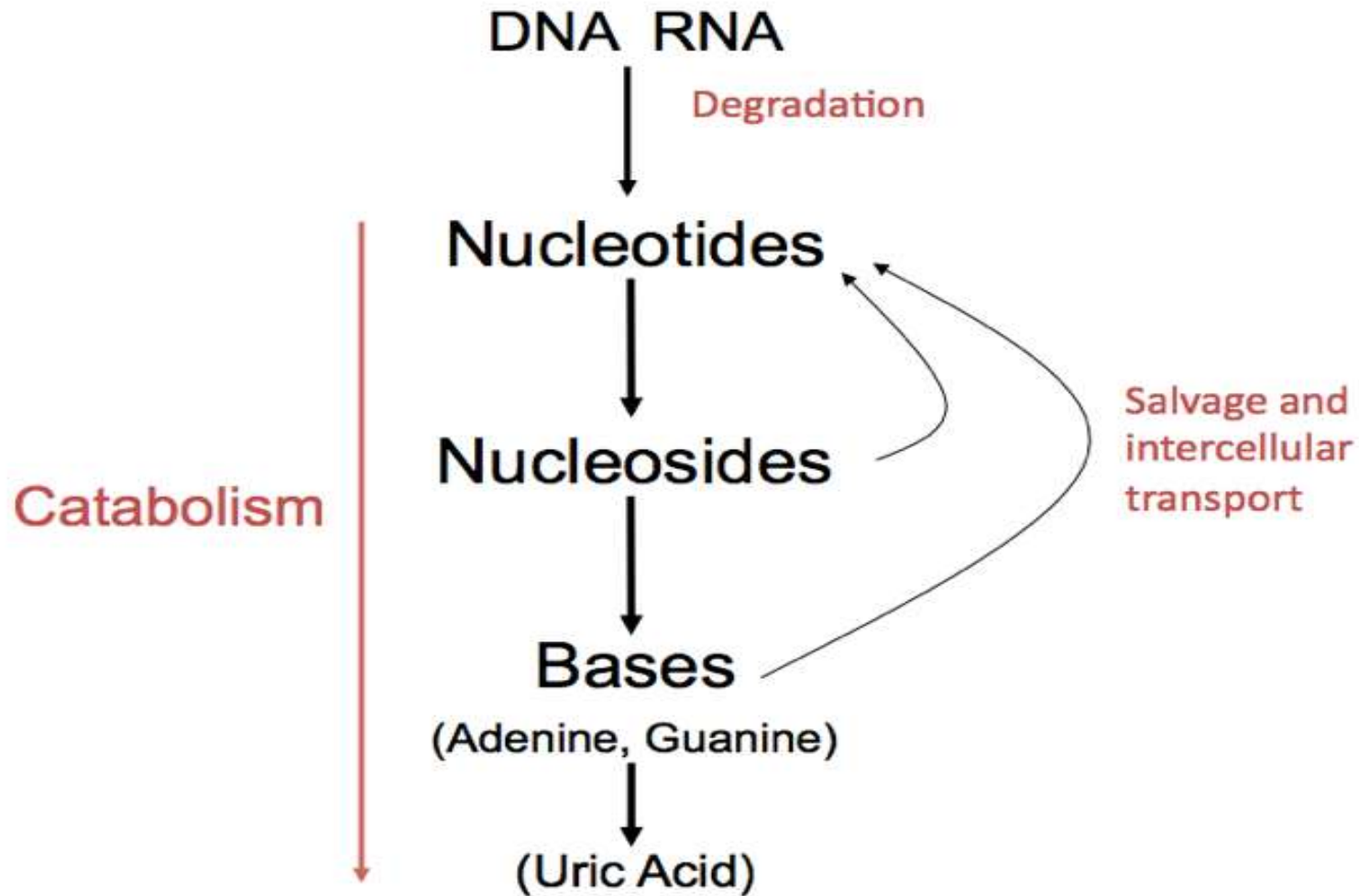


Uric Acid

NUCLEOTIDE CATABOLISM AND SALVAGE

- ❑ Nucleotide turnover occurs continuously in cells. Breakdown of DNA and RNA releases nucleoside 5'-monophosphates, which can be hydrolyzed by 5'-nucleotidases to yield nucleosides.
- ❑ Although both purine and pyrimidine nucleosides can be degraded to waste products that are excreted, the catabolic pathways have branch points in most cells at which the components of nucleotides can be salvaged.
- ❑ Having shared catabolism and salvage pathways saves metabolic energy while preventing nucleotide pools from reaching toxic levels.

NUCLEOTIDE CATABOLISM AND SALVAGE



Salvage pathway

- This pathway ensures the recycling of purines formed by degradation of nucleotides.
- Nucleosides and deoxy-nucleosides can also be salvaged.
- PRPP is the starting material in this pathway. It is also a substrate for de novo synthesis. Hence , these two pathways are closely inter-related.
- The pathway is of importance in tissues like RBCs and brain where the *de novo* pathway is not operating.

Diseases associated with purine degradation

Gout

- ❑ Gout is a disorder characterized by high levels of uric acid, the end product of purine catabolism in blood (hyperuricemia), as a result of either the overproduction or under excretion of uric acid.
- ❑ The hyperuricemia leads to the deposition of monosodium urate crystals in the joints, and an inflammatory response to the crystals, causing first acute and then to chronic gouty arthritis.

Gout-Gouty Arthritis

(a metabolic condition of abnormal purine metabolism)



Diseases associated with pyrimidine metabolism

Orotic aciduria

- ❑ Excessive excretion of **orotic acid** in urine. It causes a characteristic form of anemia and may be associated with mental and physical retardation..

Metabolic Function Of Nucleotides

1-Nucleotides are monomeric units of nucleic acids Deoxy ribonucleic acid (DNA), which include (dATP, dGTP, dTTP, and, dCTP). and Ribonucleic acid (RNA), which include (ATP, GTP, UTP, and, CTP).

2-Carries of high- energy groups. Phosphate groups of ATP, GTP,UTP.

3-Physiological mediates:

Cyclic AMP (cAMP) and Cyclic GMP (cGMP) are second messengers for most water-soluble hormones serve as metabolic regulators and activation of cell signaling.

4-Components of co enzymes

Coenzymes such as NAD, NADP, FADP and CoA are important constituents of cells and are involved in many metabolic pathways.

5-Activated intermediates:

The nucleotides serve as carriers of activated intermediates required for different reactions. Compound such as UDP_glucoese is a key intermediate in the synthesis of glycogen.

6- Many metabolic diseases have their etiology in nucleotide metabolism.

Deoxyribonucleic acid (DNA):

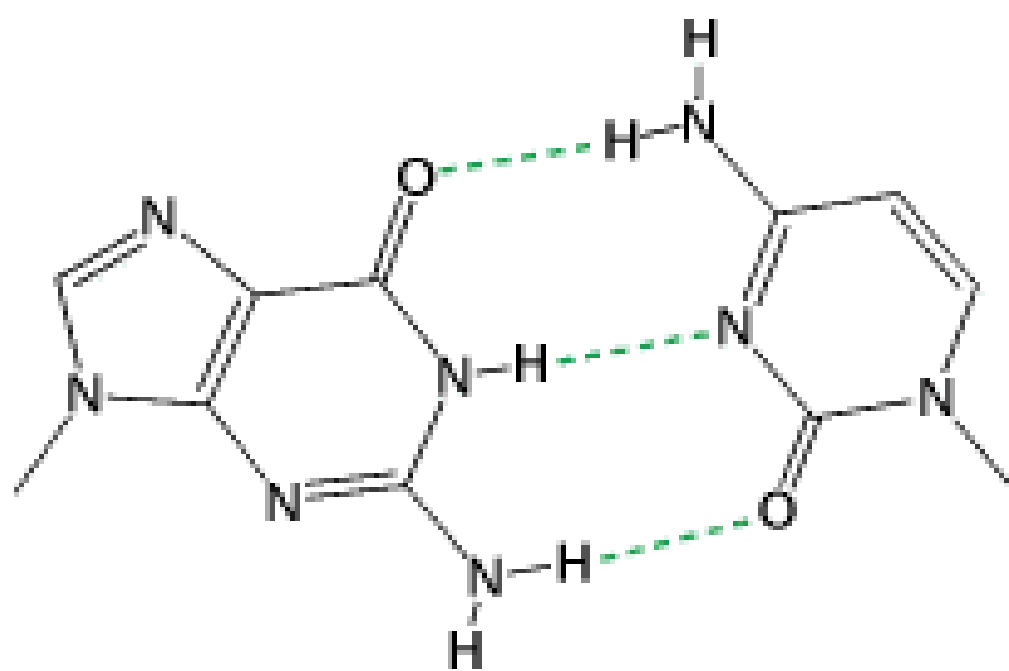
Is a polymer of deoxy ribonucleotides that are linked by 3'-5' phosphodiester bonds, the precursors of DNA are deoxyribonucleoside triphosphates (dATP, dGTP, dTTP, and, dCTP).

1-Shape: DNA is double strand helix, with strands that are antiparallel and complementary.

A- Antiparallel means that one chain runs in a 5' to 3' and the other runs in a 3' to 5' direction.

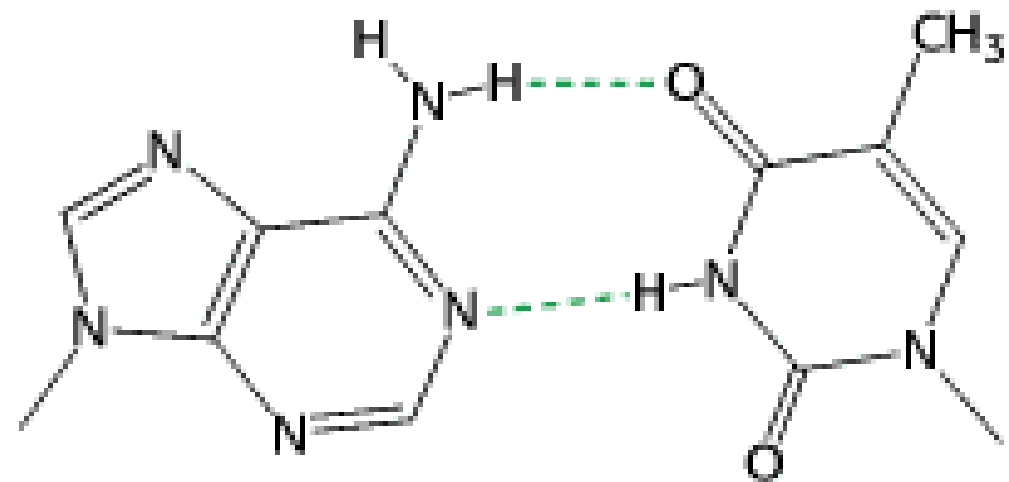
B-complementary

Means that the base adenine (A) always pairs with the base thymine (T), and the guanine (G) always pairs with the base cytosine (C). For that the number of purines always equal to the number of pyrimidines. There are 10 base pairs per turn.



Guanine

Cytosine



Adenine

Thymine

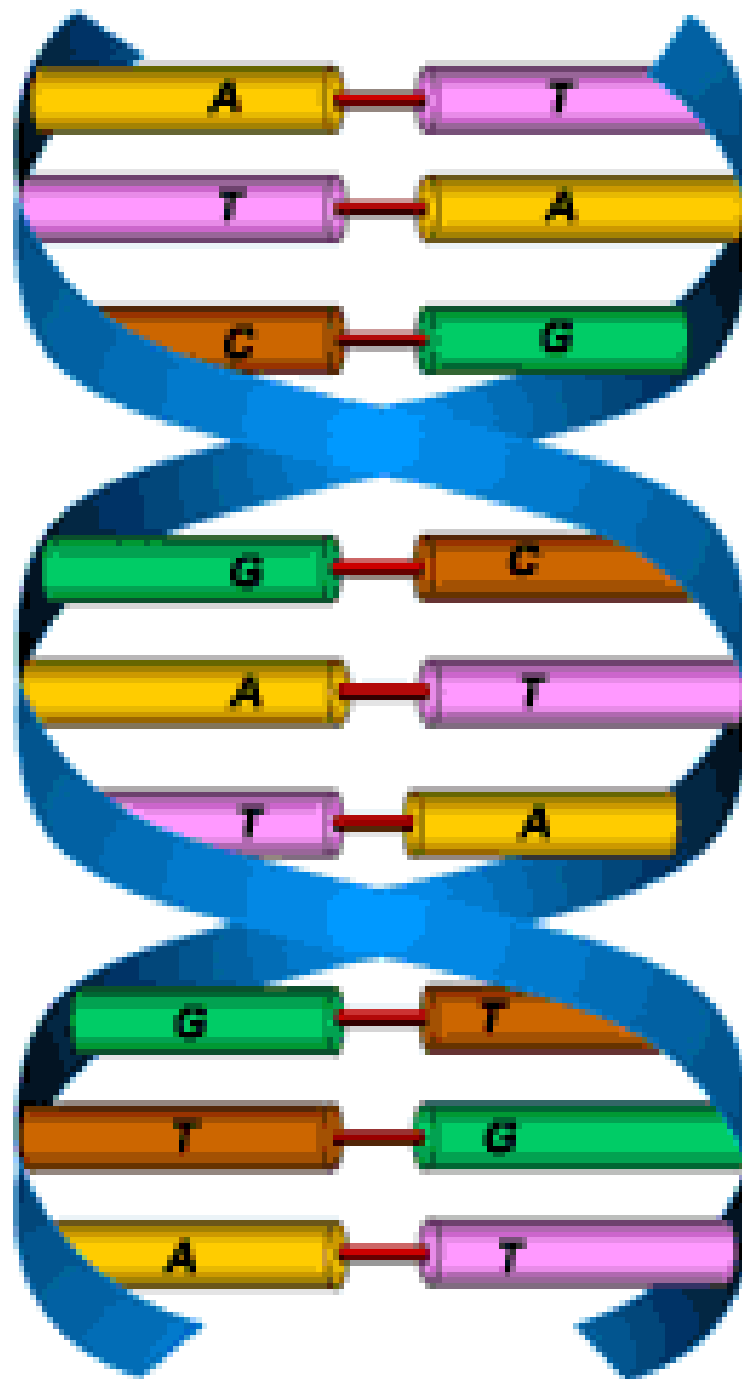
2-Stabilizing forces

The DNA double helix is stabilized to hydrogen bonds between the bases on complementary strands.

AT base pairs have two hydrogen bonds

GC base pairs have three hydrogen bonds

Back



Ribonucleic acid (RNA)

Is a polymer of ribonucleotides, is also linked by 3'-5' phosphodiester bonds.

A-RNA contains the base U instead of T as well as A, G, and C

B-Shape RNA unlike DNA, RNA is a single stranded helix.

C-There are three classes of RNA: mRNA, rRNA, tRNA

Denaturation

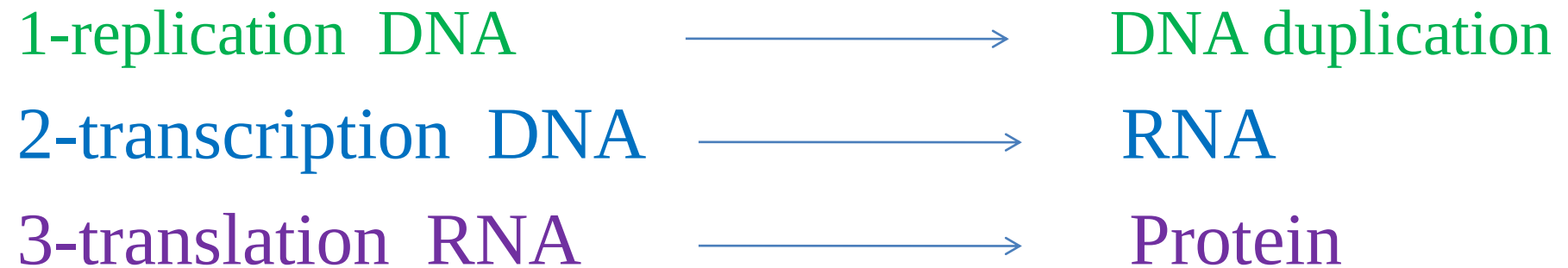
DNA and sometimes RNA unwind or denature when subjected to high T, pH extremes, and certain chemicals, eg. Formaldehyde, urea, a polynucleotide denatures at a certain temperature, known as melting temperature T_m , GC-rich regions form more stable helices than AT-rich regions, thus GC rich DNA has a higher T_m than AT rich DNA.

GENETIC INFORMATION

DNA contains genetic information. The genetic code describes the relationship between the polynucleotide alphabet of four bases and the twenty amino acids. The base sequences in one strand of parental DNA dictate the amino acid sequences of proteins.

Protein synthesis

Protein synthesis is an expression of genetic information, the making of proteins involves three processes.



1-DNA synthesis (Replication)

Is the process by which an identical copy, of DNA is made. Replication occurs every time a cell divides, so that information can be preserved and transferred to offspring.

2-Transcription

Is then process by which the genetic messages contained in DNA are “read” or transcribed. The product of transcription known as messenger RNA (mRNA). Leaves the cell nucleus and carries the message to the sites of protein synthesis.

3-Translation

Is the process by which the genetic messages carried by mRNA are described and used to build proteins.

DNA

Replication



DNA

Transcription



mRNA

Translation



Protein

Genetic code

The genetic code is universal:

The rules that relate the nucleotide sequence in messenger RNA to the amino acid sequence in protein are the same in all organisms, with exceptions in mitochondria.

* A sense code, a three nucleotide code specifies each amino acid
(eg. UUU = phenylalanine,
UCU = serine)

Some properties of genetic code

A-It is contiguous (it means codes do not overlap, and they are not separated by spaces.)

B-It is degenerate (it means there is more than one code for some amino acids.)

C-It is unambiguous (it means each code specifies only one amino acid.)